

SOLIDWORKS API (NET VBA C#) — Reference

(61/337): ICurve (part 1 of 2)

SOLIDWORKS API Documentation (converted from CHM)

ICurve (part 1 of 2) (Type)

SOLIDWORKS API Help

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ICurve Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurve Interface

Allows access to a curve and its parameters in their native form or in terms of b-curve data.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ICurve
```

Visual Basic (Usage)



```
Dim instance As ICurve
```

C#

```
public interface ICurve
```

C++/CLI

```
public interface class ICurve
```

Visual Basic for Applications (VBA) Syntax

See [Curve](#).

Example

[Get P-Spline Parameterization Data \(C#\)](#)
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[IBody2::AddProfileLine](#) and [IBody2::IAddProfileLine](#)
[IBody2::IAddProfileArcDLL](#)
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[IModeler::MergeCurves](#) and [IModeler::IMergeCurves](#)
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[ISketchPatternFeatureData::ReferencePoint](#)
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 Access Diagram

[Curve](#)

 See Also

[ICurve Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ICurve Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurve Interface

The following tables list the members exposed by [ICurve](#).

 Public Properties

Name	Description
 CircleParams	Gets the parameters of a curve that is a circle.
 ICircleParams	Gets the parameters of a curve that is a circle.
 ILineParams	Gets the parameters of a curve that is a line.
 LineParams	Gets the parameters of a curve that is a line.

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 Public Methods

Name	Description
 ApplyTransform	Applies the transform to a curve.

	ConvertArcToBcurve	Gets the b-spline value representation of the arc.
	ConvertLineToBcurve	Converts the specified line into a b-spline curve.
	Copy	Gets a copy of this curve.
	CreateTrimmedCurve	Obsolete. Superseded by ICurve::CreateTrimmedCurve2 .
	CreateTrimmedCurve2	Creates a trimmed curve.
	CreateWireBody	Creates a wire body from this curve.
	Evaluate	Obsolete. Superseded by ICurve::Evaluate2 .
	Evaluate2	Evaluates the curve at the specified parameter of the curve.
	ExtentCurve	Extends a b-spline curve by the specified length.
	FindMinimumRadius	Finds the minimum radius of curvature of the selected curve and its position and u-v parameters.
	GetBaseCurve	Gets the base curve for this trimmed curve.
	GetBCurveParams	Obsolete. Superseded by ICurve::GetBCurveParams3 .
	GetBCurveParams3	Obsolete. Superseded by ICurve::GetBCurveParams4 .
	GetBCurveParams4	Obsolete. Superseded by ICurve::GetBCurveParams5 .
	GetBCurveParams5	Gets a data object containing the parameters of a BÃ©zier curve.
	GetClosestPointOn	Determines the closest point on the curve using the x,y,z input point.
	GetEllipseParams	Gets the parameters for this elliptical curve.
	GetEndParams	Gets the end conditions of this curve.
	GetLength	Obsolete. Superseded by ICurve::GetLength2 .
	GetLength2	Obsolete. Superseded by ICurve::GetLength3 .
	GetLength3	Gets the length of a curve between the specified parameters.
	GetPCurveParams	Obsolete. Superseded by ICurve::GetPCurveParams2 .
	GetPCurveParams2	Gets the piecewise polynomial parameterization data for this curve.
	GetSplinePts	Gets the spline points for this curve.
	GetTessPts	Gets a set of points that represent the tessellation of this curve.
	IConvertArcToBcurveSize	Gets the b-curve size for the arc's conversion given the coordinates of the two end points of a line.
	IConvertLineToBcurveSize	Converts the specified line into a b-spline curve.
	IConvertPCurveToBcurveSize	Creates a b-curve from piecewise data.
	ICopy	Gets a copy of this curve.
	ICreateTrimmedCurve	Obsolete. Superseded by ICurve::CreateTrimmedCurve2 .
	Identity	Gets the type of curve.
	IEvaluate	Obsolete. Superseded by ICurve::Evaluate2 .
	IEvaluate2	Evaluates the curve at the specified parameter of the curve.
	IFindMinimumRadius	Finds the minimum radius of curvature of the selected curve and its position and u-v parameters.
	IGetBCurveParams	Obsolete. Superseded by ICurve::IGetBCurveParams3 .
	IGetBCurveParams3	Obsolete. Superseded by ICurve::GetBCurveParams4 .
	IGetBCurveParamsSize	Obsolete. Superseded by ICurve::IGetBCurveParamsSize2 .
	IGetBCurveParamsSize2	Obsolete. Superseded by ICurve::IGetBCurveParamsSize3 .
	IGetBCurveParamsSize3	Gets the b-curve size.
	IGetClosestPointOn	Determines the closest point on the curve using the x,y,z input point.
	IGetEllipseParams	Gets the parameters for this elliptical curve.
	IGetPCurveParams	Converts a curve to a piecewise rational cubic polynomial form.
	IGetPCurveParamsSize	Gets the p-curve size.

		
	IGetSplinePts	Gets the spline points for this curve.
	IGetSplinePtsSize	Gets the size of the array required by ICurve.IGetSplinePts .
	IGetTessPts	Gets a set of points that represent the tessellation of this curve.
	IGetTessPtsSize	Gets the size of the array required by ICurve.IGetTessPts .
	IntersectCurve	Gets a set of points that represent the intersection of two trimmed curves.
	IntersectCurveSize	Gets the size of the array required by ICurve.IntersectCurve .
	JoinCurves	Joins the specified curves.
	IntersectCurve	Gets a set of points that represent the intersection of two trimmed curves.
	IReverseCurve	Gets the reversed copy of this curve.
	IsBcurve	Gets whether the curve is a b-spline curve.
	IsCircle	Gets whether the curve is a circle.
	IsEllipse	Gets whether the curve is an ellipse.
	IsLine	Gets whether the curve is a line.
	IsTrimmedCurve	Gets whether the curve is trimmed.
	JoinCurves	Joins the specified curves.
	MakeBsplineCurve	Obsolete. Superseded by ICurve.MakeBsplineCurve2 .
	MakeBsplineCurve2	Creates a b-spline curve.
	ReverseCurve	Gets the reversed copy of this curve.
	ReverseEvaluate	Gets the U parameter for the given XYZ location on this curve.
	SimplifyBCurve	Simplifies a b-curve.

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See Also

[ICurve Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ICurve Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurve Interface

For a list of all members of this type, see [ICurve members](#).

Public Methods

Name	Description
 ApplyTransform	Applies the transform to a curve.
 ConvertArcToBcurve	Gets the b-spline value representation of the arc.
 ConvertLineToBcurve	Converts the specified line into a b-spline curve.
 Copy	Gets a copy of this curve.
 CreateTrimmedCurve	Obsolete. Superseded by ICurve.CreateTrimmedCurve2 .
 CreateTrimmedCurve2	Creates a trimmed curve.
 CreateWireBody	Creates a wire body from this curve.
 Evaluate	Obsolete. Superseded by ICurve.Evaluate2 .
 Evaluate2	Evaluates the curve at the specified parameter of the curve.
 ExtentCurve	Extends a b-spline curve by the specified length.

	FindMinimumRadius	Finds the minimum radius of curvature of the selected curve and its position and u-v parameters.
	GetBaseCurve	Gets the base curve for this trimmed curve.
	GetBCurveParams	Obsolete. Superseded by ICurve::GetBCurveParams3 .
	GetBCurveParams3	Obsolete. Superseded by ICurve::GetBCurveParams4 .
	GetBCurveParams4	Obsolete. Superseded by ICurve::GetBCurveParams5 .
	GetBCurveParams5	Gets a data object containing the parameters of a BÃ©zier curve.
	GetClosestPointOn	Determines the closest point on the curve using the x,y,z input point.
	GetEllipseParams	Gets the parameters for this elliptical curve.
	GetEndParams	Gets the end conditions of this curve.
	GetLength	Obsolete. Superseded by ICurve::GetLength2 .
	GetLength2	Obsolete. Superseded by ICurve::GetLength3 .
	GetLength3	Gets the length of a curve between the specified parameters.
	GetPCurveParams	Obsolete. Superseded by ICurve::GetPCurveParams2 .
	GetPCurveParams2	Gets the piecewise polynomial parameterization data for this curve.
	GetSplinePts	Gets the spline points for this curve.
	GetTessPts	Gets a set of points that represent the tessellation of this curve.
	IConvertArcToBcurveSize	Gets the b-curve size for the arc's conversion given the coordinates of the two end points of a line.
	IConvertLineToBcurveSize	Converts the specified line into a b-spline curve.
	IConvertPCurveToBcurveSize	Creates a b-curve from piecewise data.
	ICopy	Gets a copy of this curve.
	ICreateTrimmedCurve	Obsolete. Superseded by ICurve::CreateTrimmedCurve2 .
	Identity	Gets the type of curve.
	IEvaluate	Obsolete. Superseded by ICurve::Evaluate2 .
	IEvaluate2	Evaluates the curve at the specified parameter of the curve.
	IFindMinimumRadius	Finds the minimum radius of curvature of the selected curve and its position and u-v parameters.
	IGetBCurveParams	Obsolete. Superseded by ICurve::IGetBCurveParams3 .
	IGetBCurveParams3	Obsolete. Superseded by ICurve::IGetBCurveParams4 .
	IGetBCurveParamsSize	Obsolete. Superseded by ICurve::IGetBCurveParamsSize2 .
	IGetBCurveParamsSize2	Obsolete. Superseded by ICurve::IGetBCurveParamsSize3 .
	IGetBCurveParamsSize3	Gets the b-curve size.
	IGetClosestPointOn	Determines the closest point on the curve using the x,y,z input point.
	IGetEllipseParams	Gets the parameters for this elliptical curve.
	IGetPCurveParams	Converts a curve to a piecewise rational cubic polynomial form.
	IGetPCurveParamsSize	Gets the p-curve size.
	IGetSplinePts	Gets the spline points for this curve.
	IGetSplinePtsSize	Gets the size of the array required by ICurve::IGetSplinePts .
	IGetTessPts	Gets a set of points that represent the tessellation of this curve.
	IGetTessPtsSize	Gets the size of the array required by ICurve::IGetTessPts .
	IntersectCurve	Gets a set of points that represent the intersection of two trimmed curves.
	IntersectCurveSize	Gets the size of the array required by ICurve::IntersectCurve .
	JoinCurves	Joins the specified curves.
	IntersectCurve	Gets a set of points that represent the intersection of two trimmed curves.
	IReverseCurve	Gets the reversed copy of this curve.

		
	IsBcurve	Gets whether the curve is a b-spline curve.
	IsCircle	Gets whether the curve is a circle.
	IsEllipse	Gets whether the curve is an ellipse.
	IsLine	Gets whether the curve is a line.
	IsTrimmedCurve	Gets whether the curve is trimmed.
	JoinCurves	Joins the specified curves.
	MakeBsplineCurve	Obsolete, Superseded by ICurve.MakeBsplineCurve2
	MakeBsplineCurve2	Creates a b-spline curve.
	ReverseCurve	Gets the reversed copy of this curve.
	ReverseEvaluate	Gets the U parameter for the given XYZ location on this curve.
	SimplifyBCurve	Simplifies a b-curve.

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 See Also

[ICurve Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ICurve Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurve Interface

For a list of all members of this type, see [ICurve members](#).

 Public Properties

Name	Description
 CircleParams	Gets the parameters of a curve that is a circle.
 ICircleParams	Gets the parameters of a curve that is a circle.
 ILineParams	Gets the parameters of a curve that is a line.
 LineParams	Gets the parameters of a curve that is a line.

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 See Also

[ICurve Interface](#)
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SOLIDWORKS API Help

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ApplyTransform Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : ApplyTransform Method (ICurve)

Applies the transform to a curve.

 .NET Syntax

Visual Basic (Declaration)

```
Sub ApplyTransform( _  
    ByVal Xform As MathTransform _  
)
```

Visual Basic (Usage)



```
Dim instance As ICurve
Dim Xform As MathTransform

instance.ApplyTransform(Xform)
```

C#

```
void ApplyTransform(
    MathTransform Xform
)
```

C++/CLI

```
void ApplyTransform(
    & MathTransform^ Xform
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xform
Pointer to [IMathTransform](#) object

Visual Basic for Applications (VBA) Syntax

See [Curve::ApplyTransform](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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CircleParams Property (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : CircleParams Property (ICurve)

Gets the parameters of a curve that is a circle.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property CircleParams As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve
Dim value As System.Object

value = instance.CircleParams
```

C#

```
System.object CircleParams {get;}
```

C++/CLI

```
property System.Object^ CircleParams {
    System.Object^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::CircleParams](#).

Example

[Get Entities Attached to Cosmetic Thread \(VBA\)](#)

Remarks

To determine if a circular edge is a complete circle or an arc, use [IEdge::GetCurveParams2](#).

The return value is the following array of 7 double values:

[center.x, center.y, center.z, axis.x, axis.y, axis.z, radius]

Center and radius values are in meters.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::IsCircle Method](#)
[ICurve::CircleParams Property](#)
[ICurve::Identity Method](#)

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ConvertArcToBcurve Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : ConvertArcToBcurve Method (ICurve)

Gets the b-spline value representation of the arc.

.NET Syntax

Visual Basic (Declaration)

```
Function ConvertArcToBcurve( _  
    ByVal Center As System.Object, _  
    ByVal Axis As System.Object, _  
    ByVal Start As System.Object, _  
    ByVal End As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim Center As System.Object  
Dim Axis As System.Object  
Dim Start As System.Object  
Dim End As System.Object  
Dim value As System.Object  
  
value = instance.ConvertArcToBcurve(Center, Axis, Start, End)
```

C#

```
System.object ConvertArcToBcurve(  
    System.object Center,  
    System.object Axis,  
    System.object Start,  
    System.object End  
)
```

C++/CLI

```
System.Object^ ConvertArcToBcurve(
& System.Object^ Center,
& System.Object^ Axis,
& System.Object^ Start,
& System.Object^ End
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array for the x, y, z coordinates of the center point of the arc
Axis
Array for the normal vector of the arc
Start
Array for the x, y, z coordinates of the start point of the arc
End
Array for the x, y, z coordinates of the end point of the arc

Return Value

Array for the b-spline representation of the arc

 Visual Basic for Applications (VBA) Syntax

See [Curve::ConvertArcToBcurve](#).

 Remarks

Given coordinates of the center, start, end points and the normal vector of an arc, this method returns the b-spline representation of the arc. The returned data are doubles arranged in the following order:

<i>dim, order</i>	Two integers packed into a double.
<i>dim</i>	Dimension of the control point data. If dim=3 (non-rational), control point coordinates are returned as [x0,y0,z0,x1,y1,z1,.....]. If dim=4 (rational), control point coordinates are returned as [x0,y0,z0,w0,x1,y1,z1,w1,.....], where w is the weight for the point.
<i>order</i>	Order of the b-spline returned.
<i>NumControlPoints, Periodic</i>	Two integers packed into a double.
<i>NumControlPoints</i>	Number of control points for the b-spline.
<i>Periodic</i>	True if the b-spline is periodic.
<i>Knots</i>	Knot vectors. The Knot array contains (<i>NumControlPoints</i> + <i>order</i>) knot values. If the original curve is periodic, then the data is returned in a non-periodic form with extra knots inserted at the curve ends.
<i>ControlPoints</i>	List of control points output in (<i>NumControlPoints</i>) vectors of dimension dim. For non-rational b-spline, control point coordinates are output as [x0,y0,z0,x1,y1,z1,.....]. For rational b-spline, control point coordinates and the weight are output as [x0,y0,z0,w0,x1,y1,z1,w1,.....], where w is the weight for the point.

 See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::GetBCurveParams Method](#)
[ICurve::IGetBCurveParams Method](#)
[ICurve::MakeBsplineCurve Method](#)



Converts the specified line into a b-spline curve.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function ConvertLineToBcurve( _  
    ByVal StartPoint As System.Object, _  
    ByVal EndPoint As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim StartPoint As System.Object  
Dim EndPoint As System.Object  
Dim value As System.Object  
  
value = instance.ConvertLineToBcurve(StartPoint, EndPoint)
```

C#

```
System.object ConvertLineToBcurve(  
    System.object StartPoint,  
    System.object EndPoint  
)
```

C++/CLI

```
System.Object^ ConvertLineToBcurve(  
& System.Object^ StartPoint,  
& System.Object^ EndPoint  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StartPoint

Array values for the coordinates of the start point of the line

EndPoint

Array values for the coordinates of the end point of the line

Return Value

Array values giving the b-spline representation of the line

Visual Basic for Applications (VBA) Syntax

See [Curve::ConvertLineToBcurve](#).

Remarks

Given coordinates of two end points of a line, this method returns the b-spline representation of the line. The returned data are doubles arranged in the following order:

<i>dim, order</i>	Two integers packed into a double.
<i>dim</i>	Dimension of the control point data. If dim=3 (non-rational), control point coordinates are returned as [x0,y0,z0,x1,y1,z1,...]. If dim=4 (rational), control point coordinates are returned as [x0,y0,z0,w0,x1,y1,z1,w1,...], where w is the weight for the point.
<i>order</i>	Order of the b-spline returned.
<i>NumControlPoints, Periodic</i>	Two integers packed into a double.

• <i>NumControlPoints</i>	Number of control points for the b-spline.
• <i>Periodic</i>	True if the b-spline is periodic.
• <i>Knots</i>	Knot vectors. The Knot array contains (<i>NumControlPoints</i> + <i>order</i>) knot values. If the original curve is periodic, then the data is returned in a non-periodic form with extra knots inserted at the curve ends.
• <i>ControlPoints</i>	List of control points output in (<i>NumControlPoints</i>) vectors of dimension dim. For non-rational b-spline, control point coordinates are output as [x0,y0,z0,x1,y1,z1,.....]. For rational b-spline, control point coordinates and the weight are output as [x0,y0,z0,w0,x1,y1,z1,w1,.....], where w is the weight for the point.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::GetBCurveParams Method](#)
[ICurve::IGetBCurveParams Method](#)
[ICurve::IsLine Method](#)
[ICurve::MakeBsplineCurve Method](#)

SOLIDWORKS API Help

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Copy Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : Copy Method (ICurve)

Gets a copy of this curve.

.NET Syntax

Visual Basic (Declaration)

```
Function Copy() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim value As System.Object  
  
value = instance.Copy()
```

C#

```
System.object Copy()
```

C++/CLI

```
System.Object^ Copy();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Copied curve

Visual Basic for Applications (VBA) Syntax

See [Curve::Copy](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ICopy Method](#)

SOLIDWORKS API Help

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CreateTrimmedCurve Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : CreateTrimmedCurve Method (ICurve)

Obsolete. Superseded by [ICurve::CreateTrimmedCurve2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateTrimmedCurve( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim value As System.Object  
  
value = instance.CreateTrimmedCurve(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
System.object CreateTrimmedCurve(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,  
    System.double Z2  
)
```

C++/CLI

```
System.Object^ CreateTrimmedCurve(  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1

Y1

Z1

X2

Y2

Z2

Visual Basic for Applications (VBA) Syntax

See [Curve::CreateTrimmedCurve](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)

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CreateTrimmedCurve2 Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : CreateTrimmedCurve2 Method (ICurve)

Creates a trimmed curve.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateTrimmedCurve2( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
) As Curve
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim value As Curve  
  
value = instance.CreateTrimmedCurve2(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
Curve CreateTrimmedCurve2(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,  
    System.double Z2  
)
```

C++/CLI

```
Curve^ CreateTrimmedCurve2(  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X1](#)
x start point
[Y1](#)
y start point
[Z1](#)
z end point
[X2](#)

x end point

y2

y end point

z2

z end point

Return Value

Pointer to the [ICurve](#) object (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::CreateTrimmedCurve2](#).

Example

[Create Temporary Extruded Body \(VBA\)](#)
[Create Temporary Elliptical Extrusion \(VBA\)](#)
[Create Temporary Elliptical Extrusion \(VB.NET\)](#)
[Create Temporary Elliptical Extrusion \(C#\)](#)
[Create Trimmed Curve \(VBA\)](#)
[Create Trimmed Curve \(VB.NET\)](#)
[Create Trimmed Curve \(C#\)](#)

Remarks

This method trims the curve from the start point to the end point in a parametric direction. It returns NULL if the curve is discontinued or non-linear between the points.

If your application calls [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#) to create a sheet with a boundary defined by arcs ([IModeler::CreateArc](#) or [IModeler::ICreateArc](#)) or lines ([IModeler::CreateLine](#) or [IModeler::ICreateLine](#)), then your application must first call this method to trim the arc and line curves before passing them to [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#).

If your application calls [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#) to create a sheet with an elliptical boundary ([IModeler::CreateEllipse](#) or [IModeler::ICreateEllipse](#)), then you do not need to trim the elliptical curve before passing it to [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ExtentCurve Method](#)
[ICurve::ConvertPCurveToBCurveSize Method](#)
[ICurve::IsTrimmedCurve Method](#)
[ICurve::JoinCurves Method](#)
[ICurve::ReverseCurve Method](#)
[ICurve::SimplifyBCurve Method](#)
[ICurve::JoinCurves Method](#)

Availability

SOLIDWORKS 2003 SP5, Revision Number 11.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateWireBody Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : CreateWireBody Method (ICurve)

Creates a wire body from this curve.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateWireBody() As Body2
```

Visual Basic (Usage)



Copy Code

```
Dim instance As ICurve
Dim value As Body2

value = instance.CreateWireBody()
```

C#

```
Body2 CreateWireBody()
```

C++/CLI

```
Body2^ CreateWireBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the newly [body](#).

Visual Basic for Applications (VBA) Syntax

See [Curve::CreateWireBody](#).

Remarks

Wire bodies contain wires, loops, coedges, edges, and vertices. Wires typically represent profiles, construction lines, and center lines of swept shapes. Wires can also represent wire frames that, when surfaced, form shells.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[IEdge::CreateWireBody Method](#)
[IImportedCurveFeatureData::SetBody Method](#)
[IModeler::CreateWireBody Method](#)
[IModeler::ICreateWireBody Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Evaluate Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : Evaluate Method (ICurve)

Obsolete. Superseded by [ICurve::Evaluate2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Evaluate( _
    ByVal Parameter As System.Double _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve
Dim Parameter As System.Double
Dim value As System.Object

value = instance.Evaluate(Parameter)
```

C#

```
System.object Evaluate(
    System.double Parameter
```

)

C++/CLI

```
System.Object^ Evaluate(  
& System.double Parameter  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter

Curve parameter

Return Value

Array containing an array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::Evaluate](#).

Remarks

To determine a valid parameter range, use [ICurve::GetEndParams](#) or [IEdge::GetCurveParams2](#) or [IEdge::IGetCurveParams2](#).

The OLE Automation return value is an array of doubles with the following format:

[*PointX*, *PointY*, *PointZ*, *TangentX*, *TangentY*, *TangentZ*, *Success*]

where:

- *PointX*, *PointY*, and *PointZ* represent the 3D point in space for the given parameter
- *TangentX*, *TangentY*, and *TangentZ* represent the tangent vector at the point
- True if the operation is successful

This method returns values in meters.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::IEvaluate Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Evaluate2 Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : Evaluate2 Method (ICurve)

Evaluates the curve at the specified parameter of the curve.

.NET Syntax

Visual Basic (Declaration)

```
Function Evaluate2( _  
    ByVal Parameter As System.Double, _  
    ByVal NumberOfDerivatives As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As ICurve  
Dim Parameter As System.Double  
Dim NumberOfDerivatives As System.Integer
```



```
Dim value As System.Object

value = instance.Evaluate2(Parameter, NumberOfDerivatives)
```

C#

```
System.object Evaluate2(
    System.double Parameter,
    System.int NumberOfDerivatives
)
```

C++/CLI

```
System.Object^ Evaluate2(
    & System.double Parameter,
    & System.int NumberOfDerivatives
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter

Curve parameter U value (see **Remarks**)

NumberOfDerivatives

Number of derivatives (see **Remarks**)

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::Evaluate2](#).

Example

1. Open a part in SOLIDWORKS.
2. Open an Immediate window.
3. Select an edge.
4. Run this VBA macro:

```
'=====
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim boolstatus As BooleanDim longstatus As Long, longwarnings As Long
```

```
Type DoubleRec
    dValue As Double '64 bits, 8 bytes
End Type
```

```
Type Int2Rec
    iLower As Long '4 bytes
    iUpper As Long '4 bytes
End Type
```

```
' Unpack the status double
Function DoubleToLong(ByVal dVal As Double) As Long
    Dim dr As DoubleRec
    Dim i2r As Int2Rec
    ' Set the double value
    dr.dValue = dVal
    ' Copy the values
    LSet i2r = dr
    ' Extract the status value
    DoubleToLong = i2r.iLower
End Function
```

```
Sub main()
```

```

Dim status As Long
Set swApp = Application.SldWorks
Set Part = swApp.ActiveDoc
Dim myModelView As SldWorks.ModelView
Set myModelView = Part.ActiveView
myModelView.FrameState = swWindowState_e.swWindowMaximized
Dim swSelectMgr As SldWorks.SelectionMgr
Set swSelectMgr = Part.SelectionManager
Dim swEdge As SldWorks.Edge
Set swEdge = swSelectMgr.GetSelectedObject6(1, -1)
Dim swCurve As SldWorks.Curve
Dim swCurveParamData As SldWorks.CurveParamData
Set swCurve = swEdge.GetCurve
Set swCurveParamData = swEdge.GetCurveParams3
Debug.Print "The maximum U parameter value is: " & swCurveParamData.UMaxValue
Debug.Print "The minimum U parameter value is: " & swCurveParamData.UMinValue
Dim pt As Variant
pt = swCurve.Evaluate2(swCurveParamData.UMaxValue, 0)
status = DoubleToLong(pt(UBound(pt)))
If status Then
    Debug.Print " The evaluated point at UMax is:  = " & pt(0) & ", " & pt(1) & ", " & pt(2)
Else
    Debug.Print "***** Error!"
End If

End Sub
'=====

```

Example

[Create Trimmed Curve \(VBA\)](#)
[Create Trimmed Curve \(VB.NET\)](#)
[Create Trimmed Curve \(C#\)](#)

Remarks

To determine the valid range of U parameter values for Parameter, use [ICurve::GetEndParams](#) or [IEdge::GetCurveParams3](#).

The returned array contains ((NumberOfDerivatives + 1) * 3) + 1 doubles:

[*evaluated_point1*], [*evaluated_derivative_1*],...[*evaluated_derivative_NumberOfDerivatives*], [*status_code*]

where *status_code* is a packed double. After unpacking *status_code* into two longs, the lower long value is 1 for success or 0 for error. See the Example.

In terms of the number of derivatives that can be returned for a curve type:

If curve type ICurve::Identity is...	Then the maximum number of derivatives is...
Line/circle/ellipse	2
Intersection curve	2
Constant parameter line	Determined by underlying surface
SP-curve	2
B-curve	Any number

For a curve of type `swCurveTypes_e::TRIMMED_TYPE`, the number of derivatives is determined by the base curve as obtained from [ICurve::GetBaseCurve](#).

See Also

Availability

SOLIDWORKS 2007 SP5, revision Number 15.5

SOLIDWORKS API Help

[Send comments](#) on this topic.

ExtentCurve Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : ExtentCurve Method (ICurve)

Extends a b-spline curve by the specified length.

.NET Syntax

Visual Basic (Declaration)

```
Function ExtentCurve( _  
    ByVal AtStart As System.Boolean, _  
    ByVal Length As System.Double, _  
    ByVal LinearExt As System.Boolean _  
) As Curve
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim AtStart As System.Boolean  
Dim Length As System.Double  
Dim LinearExt As System.Boolean  
Dim value As Curve
```

```
value = instance.ExtentCurve(AtStart, Length, LinearExt)
```

C#

```
Curve ExtentCurve(  
    System.bool AtStart,  
    System.double Length,  
    System.bool LinearExt  
)
```

C++/CLI

```
Curve^ ExtentCurve(  
& System.bool AtStart,  
& System.double Length,  
& System.bool LinearExt  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AtStart

True to extend b-spline curve from start point, false to extend b-spline curve from end point

Length

Length by which to extend b-spline curve

LinearExt

True if the extension is linear, false if not

Return Value

Pointer to newly extended [ICurve](#) object

Visual Basic for Applications (VBA) Syntax

See [Curve::ExtentCurve](#).

Remarks

This method only affects b-spline curves. For all other types of curves, this method does nothing.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::CreateTrimmedCurve2 Method](#)
[ICurve::IConvertPcurveToBcurveSize Method](#)
[ICurve::JoinCurves Method](#)
[ICurve::JoinCurves Method](#)
[ICurve::MakeBsplinesCurve Method](#)
[ICurve::ReverseCurve Method](#)
[ICurve::SimplifyBCurve Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FindMinimumRadius Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : FindMinimumRadius Method (ICurve)

Finds the minimum radius of curvature of the selected curve and its position and u-v parameters.

.NET Syntax

Visual Basic (Declaration)

```
Function FindMinimumRadius( _  
    ByVal Bound As System.Object, _  
    ByRef NumOfRadius As System.Integer, _  
    ByRef Radius As System.Object, _  
    ByRef Location As System.Object, _  
    ByRef Parameter As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim Bound As System.Object  
Dim NumOfRadius As System.Integer  
Dim Radius As System.Object  
Dim Location As System.Object  
Dim Parameter As System.Object  
Dim value As System.Boolean  
  
value = instance.FindMinimumRadius(Bound, NumOfRadius, Radius, Location, Parameter)
```

C#

```
System.bool FindMinimumRadius(  
    System.object Bound,  
    out System.int NumOfRadius,  
    out System.object Radius,  
    out System.object Location,  
    out System.object Parameter  
)
```

C++/CLI

```
System.bool FindMinimumRadius(  
& System.Object^ Bound,  
& [Out] System.int NumOfRadius,  
& [Out] System.Object^ Radius,  
& [Out] System.Object^ Location,  
& [Out] System.Object^ Parameter  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bound
Array of doubles containing the start and end boundaries (see **Remarks**)
NumOfRadius
Number of radius returned; can be 0 or 1
Radius
Minimum radius of curvature (see **Remarks**)
Location
Position where minimum radius curvature occurred (see **Remarks**)
Parameter
Curve parameter (see **Remarks**)

Return Value

True if operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [Curve::FindMinimumRadius](#).

Example

[Find Minimum Radius of Curve \(VBA\)](#)

Remarks

The search is confined to the portion of the selected curve lying inside of Bound.

COM returns these data types for these parameters:

- Radius: VARIANT of SafeDoubleArray
- Location: VARIANT of SafeDispatchArray of [IMathpoint](#)
- Parameter: VARIANT of SafeDoubleArray

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::FindMinimumRadius Method](#)
[ISurface::FindMinimumRadius Method](#)
[ISurface::FindMinimumRadius Method](#)
[ICurve::GetEndParams Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBaseCurve Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetBaseCurve Method (ICurve)

Gets the base curve for this trimmed curve.

.NET Syntax

Visual Basic (Declaration)

Function GetBaseCurve() As Curve

Visual Basic (Usage)



Dim instance As [ICurve](#)
Dim value As Curve

value = instance.GetBaseCurve()

C#

Curve GetBaseCurve()

C++/CLI

```
Curve^ GetBaseCurve();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [ICurve](#) object

Visual Basic for Applications (VBA) Syntax

See [Curve::GetBaseCurve](#).

Remarks

This method is a geometry function that applies only to SOLIDWORKS API applications.

See Also

[ICurve Interface](#)
[ICurve Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBCurveParams Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetBCurveParams Method (ICurve)

Obsolete. Superseded by [ICurve::GetBCurveParams3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetBCurveParams( _  
    ByVal WantCubicIn As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim WantCubicIn As System.Boolean  
Dim value As System.Object  
  
value = instance.GetBCurveParams(WantCubicIn)
```

C#

```
System.object GetBCurveParams(  
    System.bool WantCubicIn  
)
```

C++/CLI

```
System.Object^ GetBCurveParams(  
&    System.bool WantCubicIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubicIn

True returns cubic rational parameters, false does not

Return Value

Array describing the parameters of the curve (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::GetBCurveParams](#).

Example

[Get Curve Spline Points \(VBA\)](#)
[Select Edges of All Holes on Face \(C#\)](#)
[Select Edges of All Holes on Face \(VB.NET\)](#)
[Select Edges of All Holes on Face \(VBA\)](#)

Remarks

To control the accuracy of the curve data, see [IModeler::SetToleranceValue](#).

The return value is an array of doubles. The array size is (2 + numKnots + numControlPointDoubles) where numKnots is (numControlPoints + order). The array is as follows:

```
[ packedDouble1, packedDouble2, knot1, knot2,..., ControlPoint1[Dimension], ControlPoint2[Dimension],... ]
```

where:

- *packedDouble1* is an integer pair containing the *Dimension* and *Order*
- *packedDouble2* is an integer pair containing the *NumControlPoints* and *Periodicity*
- *knot1*
- *knot2*

...

- *ControlPoint1[dimension]*
- *ControlPoint2[dimension]*

...

The size of the control point array is based on the curve dimension:

- If *Dimension* = 3, then *ControlPoint* is an array of 3 doubles (x, y, z)
- If *Dimension* = 4, then *ControlPoint* is an array of 4 doubles (x, y, z, w) where w = weight

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::GetBCurveParamsSize2 Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::ConvertPcurveToBcurveSize Method](#)
[ICurve::SimplifyBcurve Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBCurveParams3 Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetBCurveParams3 Method (ICurve)

Obsolete. Superseded by [ICurve::GetBCurveParams4](#)

.NET Syntax

Visual Basic (Declaration)

```
Function GetBCurveParams3( _  
    ByVal WantCubicIn As System.Boolean, _  
    ByVal WantNRational As System.Boolean, _  
    ByVal ForceNonPeriodic As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim WantCubicIn As System.Boolean  
Dim WantNRational As System.Boolean  
Dim ForceNonPeriodic As System.Boolean  
Dim value As System.Object
```

```
value = instance.GetBCurveParams3(WantCubicIn, WantNRRational, ForceNonPeriodic)
```

C#

```
System.object GetBCurveParams3(  
    System.bool WantCubicIn,  
    System.bool WantNRRational,  
    System.bool ForceNonPeriodic  
)
```

C++/CLI

```
System.Object^ GetBCurveParams3(  
& System.bool WantCubicIn,  
& System.bool WantNRRational,  
& System.bool ForceNonPeriodic  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubicIn

True returns cubic rational parameters, false does not

WantNRRational

True returns cubic rational parameters, false does not

ForceNonPeriodic

True converts the periodic curve to nonperiodic, false does not

Return Value

Array describing the parameters of the curve

Visual Basic for Applications (VBA) Syntax

See [Curve::GetBCurveParams3](#).

Remarks

To control the accuracy of the curve data, see [IModeler::SetToleranceValue](#).

The return value is an array of doubles. The array size is (2 + numKnots + numControlPointDoubles) where numKnots is (numControlPoints + order). The array is as follows:

[*packedDouble1*, *packedDouble2*, *knot1*, *knot2*,..., *ControlPoint1[Dimension]*, *ControlPoint2[Dimension]*,...]

where:

- *packedDouble1* is an integer pair containing the *Dimension* and *Order*
- *packedDouble2* is an integer pair containing the *NumControlPoints* and *Periodicity*
- *knot1*
- *knot2*
- ...
- *ControlPoint1[dimension]*
- *ControlPoint2[dimension]*
- ...

The size of the control point array is based on the curve dimension:

- If *Dimension* = 3, then *ControlPoint* is an array of 3 doubles (x, y, z)
- If *Dimension* = 4, then *ControlPoint* is an array of 4 doubles (x, y, z, w) where w = weight

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::GetBCurveParams3 Method](#)
[ICoEdge::GetCurveParams Method](#)
[ICoEdge::IGetCurveParams Method](#)
[ICurve::CircleParams Property](#)
[ICurve::ICircleParams Property](#)

[ICurve::ILineParams Property](#)
[ICurve::LineParams Property](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::GetEllipseParams Method](#)
[ICurve::IGetEllipseParams Method](#)
[ICurve::GetPCurveParams Method](#)
[ICurve::IGetPCurveParams Method](#)
[ICurve::Identity Method](#)
[ICurve::IGetBCurveParamsSize3 Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::IsCircle Method](#)
[ICurve::IsEllipse Method](#)
[ICurve::IsLine Method](#)
[ICurve::MakeBsplineCurve2 Method](#)
[IEdge::GetCurveParams2 Method](#)
[IEdge::IGetCurveParams2 Method](#)
[IModeler::CreatePCurve Method](#)
[IModeler::ICreatePCurve Method](#)

Availability

SOLIDWORKS 2007 SP5, Revision Number 15.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBCurveParams4 Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetBCurveParams4 Method (ICurve)

Obsolete. Superseded by [ICurve::GetBCurveParams5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetBCurveParams4( _  
    ByVal WantCubicIn As System.Boolean, _  
    ByVal WantNRational As System.Boolean, _  
    ByVal ForceNonPeriodic As System.Boolean, _  
    ByVal IsClosed As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim WantCubicIn As System.Boolean  
Dim WantNRational As System.Boolean  
Dim ForceNonPeriodic As System.Boolean  
Dim IsClosed As System.Boolean  
Dim value As System.Object  
  
value = instance.GetBCurveParams4(WantCubicIn, WantNRational, ForceNonPeriodic, IsClosed)
```

C#

```
System.Object GetBCurveParams4(  
    System.bool WantCubicIn,  
    System.bool WantNRational,  
    System.bool ForceNonPeriodic,  
    System.bool IsClosed  
)
```

C++/CLI

```
System.Object^ GetBCurveParams4(  
& System.bool WantCubicIn,  
& System.bool WantNRational,  
& System.bool ForceNonPeriodic,  
& System.bool IsClosed  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubicIn

True returns cubic rational parameters, false does not

WantNRational

True returns cubic rational parameters, false does not

ForceNonPeriodic

True converts the periodic curve to nonperiodic, false does not

IsClosed

True for closed curves, false otherwise

Return Value

Array describing the parameters of the curve

Visual Basic for Applications (VBA) Syntax

See [Curve::GetBCurveParams4](#).

Example

[Get BCurve Spline Points \(VBA\)](#)
[Get BCurve Spline Points \(VB.NET\)](#)
[Get BCurve Spline Points \(C#\)](#)

Remarks

To control the accuracy of the curve data, see [IModeler::SetToleranceValue](#).

The return value is an array of doubles. The array size is (2 + numKnots + numControlPointDoubles) where numKnots is (numControlPoints + order). The array is as follows:

[*packedDouble1*, *packedDouble2*, *knot1*, *knot2*,..., *ControlPoint1[Dimension]*, *ControlPoint2[Dimension]*,...]]

where:

- *packedDouble1* is an integer pair containing the *Dimension* and *Order*
- *packedDouble2* is an integer pair containing the *NumControlPoints* and *Periodicity*
- *knot1*
- *knot2*
- ...
- *ControlPoint1[dimension]*
- *ControlPoint2[dimension]*
- ...

The size of the control point array is based on the curve dimension returned in *packedDouble1*:

- If *Dimension* = 3, then *ControlPoint* is an array of 3 doubles (x, y, z)
- If *Dimension* = 4, then *ControlPoint* is an array of 4 doubles (x, y, z, w) where w = weight

The size of the knot array (NumKnots) is based on the Order returned in *packedDouble1*, and NumControlPoints and Periodicity returned in *packedDouble2* as follows:

- if Periodicity = 1, NumKnots = NumControlPoints + 1
- if Periodicity != 1, NumKnots = NumControlPoints + Order

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::GetBCurveParams3 Method](#)
[ICoEdge::GetCurveParams Method](#)
[ICoEdge::GetCurveParams Method](#)
[ICurve::CircleParams Property](#)
[ICurve::ICircleParams Property](#)
[ICurve::ILineParams Property](#)
[ICurve::LineParams Property](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::GetEllipseParams Method](#)
[ICurve::GetEllipseParams Method](#)
[ICurve::GetPCurveParams Method](#)
[ICurve::GetPCurveParams Method](#)
[ICurve::Identity Method](#)
[ICurve::GetBCurveParamsSize3 Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::IsCircle Method](#)
[ICurve::IsEllipse Method](#)
[ICurve::IsLine Method](#)
[ICurve::MakeBsplineCurve2 Method](#)
[IEdge::GetCurveParams2 Method](#)
[IEdge::GetCurveParams2 Method](#)

Availability

SOLIDWORKS 2009 SP2, Revision Number 17.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetBCurveParams5 Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetBCurveParams5 Method (ICurve)

Gets a data object containing the parameters of a Bézier curve.

.NET Syntax

Visual Basic (Declaration)

```
Function GetBCurveParams5( _  
    ByVal WantCubicIn As System.Boolean, _  
    ByVal WantNRational As System.Boolean, _  
    ByVal ForceNonPeriodic As System.Boolean, _  
    ByVal IsClosed As System.Boolean _  
) As SplineParamData
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim WantCubicIn As System.Boolean  
Dim WantNRational As System.Boolean  
Dim ForceNonPeriodic As System.Boolean  
Dim IsClosed As System.Boolean  
Dim value As SplineParamData  
  
value = instance.GetBCurveParams5(WantCubicIn, WantNRational, ForceNonPeriodic, IsClosed)
```

C#

```
SplineParamData GetBCurveParams5(  
    System.bool WantCubicIn,  
    System.bool WantNRational,  
    System.bool ForceNonPeriodic,  
    System.bool IsClosed  
)
```

C++/CLI

```
SplineParamData^ GetBCurveParams5(  
& System.bool WantCubicIn,  
& System.bool WantNRational,  
& System.bool ForceNonPeriodic,  
& System.bool IsClosed  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubicIn

True to return cubic parameters, false to not

WantNRational

True to return non-rational parameters; false to return rational parameters

ForceNonPeriodic

True to convert the curve from periodic to nonperiodic, false to not (see **Remarks**)

IsClosed

True for a closed curve, false for an open curve (see **Remarks**)

Return Value

An [ISplineParamData](#) object (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::GetBCurveParams5](#).

Example

[Get BCurve Spline Points \(C#\)](#)
[Get BCurve Spline Points \(VB.NET\)](#)
[Get BCurve Spline Points \(VBA\)](#)
[Edit Spline \(VBA\)](#)

Remarks

To control the accuracy of the curve data, see [IModeler::SetToleranceValue](#).

If a BÃ©zier curve is closed (IsClosed = true) and periodic (ForceNonPeriodic = false), the seam at the juncture of the beginning and end of the curve is continuous and smooth. The data returned in this form contains a single knot at the juncture.

If a BÃ©zier curve is closed (IsClosed = true) and non-periodic (ForceNonPeriodic = true), the seam at the juncture is continuous, contains a cusp, and is not smooth. The data returned in this form contains multiple knots at each end of the curve.

Obsolete versions of this method automatically treated closed BÃ©zier curves as periodic. The cusp at the juncture of a closed non-periodic curve (e.g., a tear drop) was lost using these methods. This method provides more flexibility for specifying precisely which form of the curve to get to prevent unintentional data loss.

This method returns the parameterization data of the BÃ©zier curve in an ISplineParamData object instead of an array. Call the methods and properties of [ISplineParamData](#) to obtain the control points, knot points, dimension, order, and periodicity of the curve.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICoEdge::GetCurveParams Method](#)
[ICoEdge::IGetCurveParams Method](#)
[ICurve::CircleParams Property](#)
[ICurve::ICircleParams Property](#)
[ICurve::ILineParams Property](#)
[ICurve::LineParams Property](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::GetEllipseParams Method](#)
[ICurve::IGetEllipseParams Method](#)
[ICurve::GetPCurveParams2 Method](#)
[ICurve::IGetPCurveParams Method](#)
[ICurve::Identity Method](#)
[ICurve::IGetBCurveParamsSize3 Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::IsCircle Method](#)
[ICurve::IsEllipse Method](#)
[ICurve::IsLine Method](#)
[ICurve::MakeBsplineCurve2 Method](#)
[IEdge::GetCurveParams3 Method](#)
[IEdge::IGetCurveParams2 Method](#)
[IModeler::CreatePCurve Method](#)
[IModeler::ICreatePCurve Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetClosestPointOn Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetClosestPointOn Method (ICurve)

Determines the closest point on the curve using the x,y,z input point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetClosestPointOn( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As ICurve  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.GetClosestPointOn(X, Y, Z)
```

C#

```
System.object GetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ GetClosestPointOn(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value of the input point

Y
Y value of the input point

Z
Z value of the input point

Return Value

Array of 5 doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::GetClosestPointOn](#).

Remarks

This method returns only one point, regardless of the number of valid minimum distance points.

The array return contains the following values:

[*PointX*, *PointY*, *PointZ*, *ParamU*, *NotUsed*]

where:

- *PointX*, *PointY*, and *PointZ* represent the point on the curve
- *ParamU* is the parameter on the curve that is closest to the input point
- *NotUsed* is unused

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::GetClosestPointOn Method](#)
[IEdge::GetClosestPointOn Method](#)
[IEdge::IGetClosestPointOn Method](#)
[IFace2::GetClosestPointOn Method](#)
[IFace2::IGetClosestPointOn Method](#)
[ISurface::GetClosestPointOn Method](#)
[ISurface::IGetClosestPointOn Method](#)
[IVertex::GetClosestPointOn Method](#)
[IVertex::IGetClosestPointOn Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEllipseParams Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetEllipseParams Method (ICurve)

Gets the parameters for this elliptical curve.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEllipseParams() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim value As System.Object
```

```
value = instance.GetEllipseParams()
```

C#

```
System.Object GetEllipseParams()
```

C++/CLI

```
System.Object^ GetEllipseParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Object (see [Remarks](#))

Visual Basic for Applications (VBA) Syntax

See [Curve::GetEllipseParams](#).

Remarks

Use [ICurve::IsEllipse](#) to determine if a curve is an ellipse.

The return value is an array of doubles with the following format:

[*centerptX*, *centerptY*, *centerptZ*, *majorRad*, *majorAxisX*, *majorAxisY*, *majorAxisZ*, *minorRad*, *minorAxisX*, *minorAxisY*, *minorAxisZ*]

where:

- *centerptX*, *centerptY*, *centerptZ* = location of ellipse center
- *majorRad* = major radius
- *majorAxisX*, *majorAxisY*, *majorAxisZ* = major axis values
- *minorRad* = minor radius
- *minorAxisX*, *minorAxisY*, *minorAxisZ* = minor axis values

For Dispatch users, if this curve is not an ellipse, then SOLIDWORKS returns an empty VARIANT. Otherwise, SOLIDWORKS returns a SafeArray in Visual Basic for Applications (VBA).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::Identity Method](#)
[ICurve::GetEllipseParams Method](#)
[ICurve::IsEllipse Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1999042

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEndParams Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetEndParams Method (ICurve)

Gets the end conditions of this curve.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetEndParams( _  
    ByRef Start As System.Double, _  
    ByRef End As System.Double, _  
    ByRef IsClosed As System.Boolean, _  
    ByRef IsPeriodic As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim Start As System.Double  
Dim End As System.Double  
Dim IsClosed As System.Boolean  
Dim IsPeriodic As System.Boolean  
Dim value As System.Boolean  
  
value = instance.GetEndParams(Start, End, IsClosed, IsPeriodic)
```

C#

```
System.bool GetEndParams(  
    out System.double Start,  
    out System.double End,  
    out System.bool IsClosed,  
    out System.bool IsPeriodic  
)
```

C++/CLI

```
System.bool GetEndParams(  
& [Out] System.double Start,  
& [Out] System.double End,  
& [Out] System.bool IsClosed,  
& [Out] System.bool IsPeriodic  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Start
Start parameter value

End
End parameter value

IsClosed
True for closed curves, false for other types

IsPeriodic
True for periodic curves, false for other types

Return Value

True if the operation was successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Curve::GetEndParams](#).

Example

[Create Space Parameter Curve on Surface \(VBA\)](#)
[Evaluate Curves Defined in Sketch Space \(VBA\)](#)
[Find Outside Edges of Face \(VBA\)](#)
[Get Intersecting Faces \(VBA\)](#)
[Get Sketch Segment and Curve Data \(VBA\)](#)
[Get Start and End Points of Spline \(VBA\)](#)
[Create Trimmed Curve \(VBA\)](#)
[Create Trimmed Curve \(VB.NET\)](#)
[Create Trimmed Curve \(C#\)](#)

Remarks

The parametric range is the valid range for the curve. If the curve is infinite (such as a line), then the parametric maximum is the largest permissible value in the parametric space.

See Also

[ICurve Interface](#)
[ICurve Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLength Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetLength Method (ICurve)

Obsolete. Superseded by [ICurve::GetLength2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetLength( _  
    ByVal StartParam As System.Double, _  
    ByVal EndParam As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim StartParam As System.Double  
Dim EndParam As System.Double  
Dim value As System.Double  
  
value = instance.GetLength(StartParam, EndParam)
```

C#

```
System.double GetLength(  
    System.double StartParam,  
    System.double EndParam  
)
```

C++/CLI

```
System.double GetLength(  
& System.double StartParam,  
& System.double EndParam  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StartParam

EndParam

Visual Basic for Applications (VBA) Syntax

See [Curve::GetLength](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetLength2 Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetLength2 Method (ICurve)

Obsolete. Superseded by [ICurve::GetLength3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetLength2( _  
    ByVal StartParam As System.Double, _  
    ByVal EndParam As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim StartParam As System.Double  
Dim EndParam As System.Double  
Dim value As System.Double  
  
value = instance.GetLength2(StartParam, EndParam)
```

C#

```
System.double GetLength2(  
    System.double StartParam,  
    System.double EndParam  
)
```

C++/CLI

```
System.double GetLength2(  
& System.double StartParam,  
& System.double EndParam  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StartParam

Start parameter

EndParam

End parameter

Return Value

Length of the curve between the two parameters

Visual Basic for Applications (VBA) Syntax

See [Curve::GetLength2](#).

Example

[Get Edge Curve Parameterization \(VBA\)](#)
[Get intersecting Faces \(VBA\)](#)
[Get Length of Edge \(VBA\)](#)
[Get Reference Curve Information \(VBA\)](#)
[Get Start and End Points of Spline \(VBA\)](#)

See Also

[ICurve Interface](#)
[ICurve Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetLength3 Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetLength3 Method (ICurve)

Gets the length of a curve between the specified parameters.

.NET Syntax

Visual Basic (Declaration)

```
Function GetLength3( _  
    ByVal StartParam As System.Double, _  
    ByVal EndParam As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim StartParam As System.Double  
Dim EndParam As System.Double  
Dim value As System.Double  
  
value = instance.GetLength3(StartParam, EndParam)
```

C#

```
System.double GetLength3(  
    System.double StartParam,  
    System.double EndParam  
)
```

C++/CLI

```
System.double GetLength3(  
& System.double StartParam,  
& System.double EndParam  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StartParam
Start parameter
EndParam
End parameter

Return Value

Length of the curve between the two parameters

Visual Basic for Applications (VBA) Syntax

See [Curve::GetLength3](#).

Remarks

This method returns the length of the holding (e.g., trimmed) curve and not the base curve. The now obsolete [ICurve::Length2](#) returned the length of the base curve.

See Also

[ICurve Interface](#)
[ICurve Members](#)

Availability

SOLIDWORKS 2008 SP1, Revision Number 16.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPCurveParams Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetPCurveParams Method (ICurve)

Obsolete. Superseded by [ICurve::GetPCurveParams2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPCurveParams() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim value As System.Object  
  
value = instance.GetPCurveParams()
```

C#

```
System.object GetPCurveParams()
```

C++/CLI

```
System.Object^ GetPCurveParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles describing the parameters of the curve (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::GetPCurveParams](#).

Example

Remarks

This method returns the curve as a series of segments, each of which has this form:

$$p(t) = \frac{(w_0 \cdot a_0 + w_1 \cdot a_1 \cdot t + w_2 \cdot a_2 \cdot t^2 + w_3 \cdot a_3 \cdot t^3)}{(w_0 + w_1 \cdot t + w_2 \cdot t^2 + w_3 \cdot t^3)}$$

Where w is the weight coefficient and a is the polynomial coefficient.

The coefficients returned are vectors of dimensions "dim". For example, if the curve is rational, then *dimension* = 4, and the coefficients returned for each segment are as follows:

(a0x,a0y,a0z,w0),(a1x,a1y,a1z,w1)...

However, if the curve is returned as non-rational, then *dimension* = 3, and its coefficients are:

(a0x,a0y,a0z),(a1x,a1y,a1z),...

The size of the return value array is (3 + *NumberKnots* + *NumberSegments* x *Order* x *Dimension*).

The array is as follows:

[*packedDouble1*, *packedDouble2*, *packedDouble3*, *knot1*, *knot2*,..., *CoefficientsSegment1*[*Dimension*],
CoefficientsSegment2[*Dimension*],...]

where:

packedDouble1: Integer pair containing *Reserved* and *Order*

packedDouble2: Integer pair containing *NumberSegments* and *Periodicity*

packedDouble3: Integer pair containing *NumberKnots* and *Dimension*

knot1

knot2

...

CoefficientsSegment1[*Dimension*]

CoefficientsSegment2[*Dimension*]

...

The coefficients for each segment start with the constant term and end with the term of highest degree (for example, ConstantX, ConstantY, ConstantZ, AX, AY, AZ, BX, BY, BZ). The total number of coefficients is:

(*NumberSegments*) x (*Order*) x (*Dimension*).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::Identity Method](#)
[ICurve::GetPCurveParams Method](#)
[ICurve::GetPCurveParamsSize Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPCurveParams2 Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetPCurveParams2 Method (ICurve)

Gets the piecewise polynomial parameterization data for this curve.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPCurveParams2() As SplineParamData
```

Visual Basic (Usage)

```
Dim instance As ICurve  
Dim value As SplineParamData
```

```
value = instance.GetPCurveParams2()
```

C#

```
SplineParamData GetPCurveParams2()
```

C++/CLI

```
SplineParamData^ GetPCurveParams2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

An [ISplineParamData](#) object

Visual Basic for Applications (VBA) Syntax

See [Curve::GetPCurveParam2](#).

Example

[Get P-Spline Parameterization Data \(C#\)](#)
[Get P-Spline Parameterization Data \(VB.NET\)](#)
[Get P-Spline Parameterization Data \(VBA\)](#)

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::Identify Method](#)
[ICurve::IGetPCurveParams Method](#)
[ICurve::IGetPCurveParamsSize Method](#)
[ICurve::GetBCurveParams5 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSplinePts Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetSplinePts Method (ICurve)

Gets the spline points for this curve.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplinePts(  
    ByVal ParamsArrayIn As System.Object  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim ParamsArrayIn As System.Object  
Dim value As System.Object  
  
value = instance.GetSplinePts(ParamsArrayIn)
```

C#

```
System.object GetSplinePts(  
    System.object ParamsArrayIn  
)
```

C++/CLI

```
System.Object^ GetSplinePts(  
& System.Object^ ParamsArrayIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ParamsArrayIn
Array containing the definition of the spline

Return Value

Array containing the spline points

Visual Basic for Applications (VBA) Syntax

See [Curve::GetSplinePts](#).

Example

[Get Curve Spline Points \(VBA\)](#)
[Get Length of Spline or Elliptical Edge \(VBA\)](#)
[Get Parameters and Spline Points for Curve \(VBA\)](#)

Remarks

For OLE Automation applications, the ParamsIn argument contains an array defining the spline. This array contains the values for propArray, knotsArray and ctrlPntCoordArray, all packed into a single array (propArray is packed as four integers into the first two elements of the array.) The contents of these arrays is described in [ICurve::IGetSplinePtsSize](#).

The array returned contains the x, y, z location of the spline points:

[x1, y1, z1, x2, y2, z2,...]

Data passed into this method must satisfy the following requirements:

- If the curve is periodic, it must not have any repeated knots.
- If the curve is non-periodic, it must only have repeated knots at its ends.
- The curve must be cubic or lower degree, non-rational, and have continuous first and second derivatives.

NOTE: For a linear or quadratic curve to satisfy these continuity requirements, it must consist of a single segment.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::IGetSplinePts Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetTessPts Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : GetTessPts Method (ICurve)

Gets a set of points that represent the tessellation of this curve.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTessPts( _  
    ByVal ChordTolerance As System.Double, _  
    ByVal LengthTolerance As System.Double, _  
    ByVal StartPoint As System.Object, _  
    ByVal EndPoint As System.Object _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As ICurve  
Dim ChordTolerance As System.Double  
Dim LengthTolerance As System.Double  
Dim StartPoint As System.Object  
Dim EndPoint As System.Object  
Dim value As System.Object
```

```
value = instance.GetTessPts(ChordTolerance, LengthTolerance, StartPoint, EndPoint)
```

C#

```
System.object GetTessPts(  
    System.double ChordTolerance,  
    System.double LengthTolerance,  
    System.object StartPoint,
```



```
System.Object EndPoint
)
```

C++/CLI

```
System.Object^ GetTessPts(
& System.double ChordTolerance,
& System.double LengthTolerance,
& System.Object^ StartPoint,
& System.Object^ EndPoint
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ChordTolerance

Chord tolerance to use in tessellation (meters); this is the maximum permitted distance from a cord to the curve between the cord endpoints; the smallest allowable value for this parameter is 1E-8; if 0.0 or a value smaller than 1E-8 is specified, then 1E-8 is used by default

LengthTolerance

Length tolerance to be used to filter out very short segments (meters); this method does not return tessellated segments shorter than this value

StartPoint

Array containing the start point of the curve

EndPoint

Array containing the end point of the curve

Return Value

Array of doubles containing the tessellation points (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::GetTessPts](#).

Example

[Create Space Parameter Curve on Surface \(VBA\)](#)

[Find Outside Edges of Face \(VBA\)](#)

[Get Intersecting Faces \(VBA\)](#)

[Get Length of Edge \(VBA\)](#)

[Get Reference Curve Information \(VBA\)](#)

Remarks

The array returned contains the x, y, z location of the tessellation points:

[x1, y1, z1, x2, y2, z2, ...]

See Also

[ICurve Interface](#)

[ICurve Members](#)

[ICurve::IGetTessPts Method](#)

[ICurve::IGetTessPtsSize Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

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ICircleParams Property (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : ICircleParams Property (ICurve)

Gets the parameters of a curve that is a circle.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property ICircleParams As System.Double

Visual Basic (Usage)

```
Dim instance As ICurve  
Dim value As System.Double  
  
value = instance.ICircleParams
```

C#

```
System.double ICircleParams {get;}
```

C++/CLI

```
property System.double ICircleParams {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to an array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::ICircleParams](#).

Remarks

To determine if a circular edge is a complete circle or an arc, use [IEdge::IGetCurveParams2](#).

The return value is the following array of 7 double values:

[*center.x*, *center.y*, *center.z*, *axis.x*, *axis.y*, *axis.z*, *radius*]

Center and radius values are in meters.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::CircleParams Property](#)
[ICurve::IsCircle Method](#)
[ICurve::Identity Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IConvertArcToBcurveSize Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IConvertArcToBcurveSize Method (ICurve)

Gets the b-curve size for the arc's conversion given the coordinates of the two end points of a line.

.NET Syntax

Visual Basic (Declaration)

```
Function IConvertArcToBcurveSize( _  
    ByRef Center As System.Double, _  
    ByRef Axis As System.Double, _  
    ByRef Start As System.Double, _  
    ByRef End As System.Double _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ICurve  
Dim Center As System.Double  
Dim Axis As System.Double  
Dim Start As System.Double  
Dim End As System.Double  
Dim value As System.Integer
```

```
value = instance.IConvertArcToBcurveSize(Center, Axis, Start, End)
```

C#

```
System.int IConvertArcToBcurveSize(  
    ref System.double Center,  
    ref System.double Axis,  
    ref System.double Start,  
    ref System.double End  
)
```

C++/CLI

```
System.int IConvertArcToBcurveSize(  
    & System.double% Center,  
    & System.double% Axis,  
    & System.double% Start,  
    & System.double% End  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Pointer to an array of doubles (x, y, z), the coordinates of the arc center

Axis

Pointer to an array of doubles (x, y, z), the normal vector of the arc

Start

Pointer to an array of doubles (x, y, z), the coordinates of the arc start point

End

Pointer to an array of doubles (x, y, z), the coordinates of the arc end point

Return Value

Size of the data necessary for the conversion to a b-curve

Visual Basic for Applications (VBA) Syntax

See [Curve::IConvertArcToBcurveSize](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::ExtentCurve Method](#)
[ICurve::GetBCurveParams Method](#)
[ICurve::GetBCurveParams Method](#)
[ICurve::GetBCurveParamsSize2 Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::MakeBsplineCurve Method](#)
[ICurve::SimplifyBCurve Method](#)
[ICurve::IConvertLineToBcurveSize Method](#)
[ICurve::IConvertPcurveToBcurveSize Method](#)
[ICurve::ConvertLineToBcurve Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IConvertLineToBcurveSize Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IConvertLineToBcurveSize Method (ICurve)

Converts the specified line into a b-spline curve.

.NET Syntax

Visual Basic (Declaration)

```
Function IConvertLineToBcurveSize( _  
    ByRef StartPoint As System.Double, _  
    ByRef EndPoint As System.Double _  
) As System.Integer
```

```
Dim instance As ICurve
Dim StartPoint As System.Double
Dim EndPoint As System.Double
Dim value As System.Integer

value = instance.IConvertLineToBcurveSize(StartPoint, EndPoint)
```

C#

```
System.int IConvertLineToBcurveSize(
    ref System.double StartPoint,
    ref System.double EndPoint
)
```

C++/CLI

```
System.int IConvertLineToBcurveSize(
    & System.double% StartPoint,
    & System.double% EndPoint
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StartPoint

Array values for the coordinates of the start point of the line

EndPoint

Array values for the coordinates of the end point of the line

Return Value

Array giving the b-spline representation of the line

Visual Basic for Applications (VBA) Syntax

See [Curve.IConvertLineToBcurveSize](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::IsLine Method](#)
[ICurve::MakeBsplineCurve Method](#)
[ICurve::IConvertArcToBcurveSize Method](#)
[ICurve::IConvertPcurveToBcurveSize Method](#)
[ICurve::GetBCurveParams Method](#)
[ICurve::ExtendCurve Method](#)
[ICurve::IGetBCurveParams Method](#)
[ICurve::IGetBCurveParamsSize2 Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::MakeBsplineCurve Method](#)
[ICurve::SimplifyBcurve Method](#)

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IConvertPcurveToBcurveSize Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IConvertPcurveToBcurveSize Method (ICurve)

Creates a b-curve from piecewise data.

.NET Syntax

Visual Basic (Declaration)

```
Function IConvertPcurveToBcurveSize( _
    ByVal Dim As System.Integer, _
    ByVal Order As System.Integer, _
    ByVal Nsegs As System.Integer, _
    ByRef Coeffs As System.Double, _
```

```

    ByVal Basis As System.Integer, _
    ByRef Xform As System.Double, _
    ByVal ScaleFactor As System.Double _
) As System.Integer

```

Visual Basic (Usage)



```

Dim instance As ICurve
Dim Dim As System.Integer
Dim Order As System.Integer
Dim Nsegs As System.Integer
Dim Coeffs As System.Double
Dim Basis As System.Integer
Dim Xform As System.Double
Dim ScaleFactor As System.Double
Dim value As System.Integer

value = instance.IConvertPcurveToBcurveSize(Dim, Order, Nsegs, Coeffs, Basis, Xform, ScaleFactor)

```

C#

```

System.int IConvertPcurveToBcurveSize(
    System.int Dim,
    System.int Order,
    System.int Nsegs,
    ref System.double Coeffs,
    System.int Basis,
    ref System.double Xform,
    System.double ScaleFactor
)

```

C++/CLI

```

System.int IConvertPcurveToBcurveSize(
    & System.int Dim,
    & System.int Order,
    & System.int Nsegs,
    & System.double% Coeffs,
    & System.int Basis,
    & System.double% Xform,
    & System.double ScaleFactor
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Dim

Dimension of the curve:

- 3 = nonrational
- 4 = rational

Order

Order of the curve; minimum is 2 (linear)

Nsegs

Number of segments in the curve; must be at least 1

Coeffs

Array of coefficients

Basis

0; not currently used

Xform

Transformation matrix to apply after conversion to b-curve

ScaleFactor

Units conversion factor to apply after conversion to b-curve

Return Value

Indicates that conversion succeeded or failed

Remarks

Dimension of coefficient vectors dim:

- For rational curves dim = 4.
- For non-rational curves dim = 3.

Order of each segment of the curve order:

1. The order of the curve = degree + 1.
2. The minimum order is 2.

Number of segments in the curve nseg:

- There must be at least one segment (nseg >= 1).
- Adjacent segments must meet.

Coefficient data coeffs:

- Contains order * nseg vectors of dimension dim. If dim = 3, then the vectors are 3-D vectors giving the x, y and z components. If dim = 4 then each vector has a weight (w) associated with it, and x, y, z and w components are supplied for each vector. The weights supplied must be greater than 0.
- The coefficients are supplied in order, segment by segment.
- The interpretation of the coefficients depends on the representation method chosen; this is determined by the value of the argument basis.

Representation method basis:

The expressions for each segment of the B-curve P(t) in the various representations are shown next. For generality, the rational form is given.

The simplification to the non-rational form can be obtained by setting both the weights and the denominator equal to 1.0

Polynomial coefficients:

The curve equation is given by a rational polynomial of order 'order':

$$P(t) = \frac{\sum_{i=0}^n w_i A_i t^i}{\sum_{i=0}^n w_i t^i}$$

where n = order-1

A = polynomial coefficient

i

w = weight for A

i i

The polynomial coefficients are supplied starting with the constant term and ending with the term of highest degree.

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::ExtentCurve Method](#)
[ICurve::MakeBspineCurve Method](#)
[ICurve::SimplifyBCurve Method](#)
[ConvertArcToBcurve Method](#)
[ConvertLineToBcurve Method](#)
[GetBcurveParams Method](#)
[ICurve::IConvertArcToBcurveSize Method](#)
[ICurve::IConvertLineToBcurveSize Method](#)
[ICurve::IGetBcurveParams Method](#)
[ICurve::IGetBcurveParamsSize2 Method](#)
[ICurve::IsBcurve Method](#)

ICopy Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : ICopy Method (ICurve)

Gets a copy of this curve.

.NET Syntax

Visual Basic (Declaration)

```
Function ICopy() As Curve
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim value As Curve  
  
value = instance.ICopy()
```

C#

```
Curve ICopy()
```

C++/CLI

```
Curve^ ICopy();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the copied [curve](#)

Visual Basic for Applications (VBA) Syntax

See [Curve::ICopy](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::Copy Method](#)

SOLIDWORKS API Help

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ICreateTrimmedCurve Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : ICreateTrimmedCurve Method (ICurve)

Obsolete. Superseded by [ICurve::CreateTrimmedCurve2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateTrimmedCurve( _  
    ByRef Start As System.Double, _  
    ByRef End As System.Double _  
) As Curve
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim Start As System.Double  
Dim End As System.Double  
Dim value As Curve  
  
value = instance.ICreateTrimmedCurve(Start, End)
```

C#

```
Curve ICreateTrimmedCurve(  
    ref System.double Start,  
    ref System.double End  
)
```

C++/CLI

```
Curve^ ICreateTrimmedCurve(  
& System.double% Start,  
& System.double% End  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Start

End

Visual Basic for Applications (VBA) Syntax

See [Curve::ICreateTrimmedCurve](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)

SOLIDWORKS API Help

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Identity Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : Identity Method (ICurve)

Gets the type of curve.

.NET Syntax

Visual Basic (Declaration)

```
Function Identity() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim value As System.Integer  
  
value = instance.Identity()
```

C#

```
System.int Identity()
```

C++/CLI

```
System.int Identity();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of curve as defined in [swCurveTypes_e](#)

Visual Basic for Applications (VBA) Syntax

See [Curve::Identity](#).

Example

[Get Length of Edge \(VBA\)](#)
[Get Length of Spline or Elliptical Edge \(VBA\)](#)

Remarks

If the return value is TRIMMED_TYPE, then the original untrimmed curve type may be established by calling [ICurve::IsLine](#), [ICurve::IsBcurve](#) or [ICurve::IsCircle](#).

To determine if a circular edge is a complete circle or an arc, see [IEdge::GetCurveParams2](#) or [IEdge::IGetCurveParams2](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::IsEllipse Method](#)
[ICurve::IsTrimmedCurve Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::IsCircle Method](#)
[ICurve::IsLine Method](#)

SOLIDWORKS API Help

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IEvaluate Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IEvaluate Method (ICurve)

Obsolete. Superseded by [ICurve::Evaluate2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IEvaluate( _  
    ByVal Parameter As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim Parameter As System.Double  
Dim value As System.Double  
  
value = instance.IEvaluate(Parameter)
```

C#

```
System.double IEvaluate(  
    System.double Parameter  
)
```

C++/CLI

```
System.double IEvaluate(  
& System.double Parameter  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter
Curve parameter

Return Value

Pointer to an array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::IEvaluate](#).

Remarks

To determine a valid parameter range, use [ICurve::GetEndParams](#) or [IEdge::GetCurveParams2](#) or [IEdge::IGetCurveParams2](#).

The OLE Automation return value is an array of doubles with the following format:

[*PointX*, *PointY*, *PointZ*, *TangentX*, *TangentY*, *TangentZ*, *Success*]

where:

- *PointX*, *PointY*, and *PointZ* represent the 3D point in space for the given parameter
- *TangentX*, *TangentY*, and *TangentZ* represent the tangent vector at the point
- True if the operation is successful

The COM return value is an array of 6 doubles representing the point and tangent. The success value is determined from the HRESULT return.

This method returns values in meters.

See Also

[ICurve Interface](#)
[ICurve Members](#)

SOLIDWORKS API Help

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IEvaluate2 Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IEvaluate2 Method (ICurve)

Evaluates the curve at the specified parameter of the curve.

.NET Syntax

Visual Basic (Declaration)

```
Function IEvaluate2( _  
    ByVal Parameter As System.Double, _  
    ByVal NumberOfDerivatives As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurve  
Dim Parameter As System.Double  
Dim NumberOfDerivatives As System.Integer  
Dim value As System.Double  
  
value = instance.IEvaluate2(Parameter, NumberOfDerivatives)
```

C#

```
System.double IEvaluate2(  
    System.double Parameter,  
    System.int NumberOfDerivatives  
)
```

C++/CLI

```
System.double IEvaluate2(  
& System.double Parameter,  
& System.int NumberOfDerivatives  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameter

Curve parameter

NumberOfDerivatives

Number of derivatives

Return Value

Array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::IEvaluate2](#).

Remarks

To determine a valid parameter range, use [ICurve::GetEndParams](#) or [IEdge::GetCurveParams2](#).

The format of the return value is an array of (NumberOfDerivatives + 1) * 3 doubles:

[*evaluated point*], [*evaluated derivative 1*],...[*evaluated derivative NumberOfDerivatives*]

In pseudo mathematical notation, this could be written as:

P(t) P(t)/dt P(t)/dtdt

In terms of the number of derivatives that can be returned for a curve type, you could write:

Curve type	Maximum number of derivatives
Line/circle/ellipse	2
Intersection curve	2
Constant parameter line	Determined by underlying surface
SP-curve	2
B-curve	Any number

where the curve type is from [ICurve::Identity](#).

For a curve of type swCurveTypes_e::TRIMMED_TYPE, the number of derivatives is determined by the base curve as obtained from [ICurve::GetBaseCurve](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::Evaluate2 Method](#)

Availability

SOLIDWORKS 2007 SP5, Revision Number 15.5

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IFindMinimumRadius Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IFindMinimumRadius Method (ICurve)

Finds the minimum radius of curvature of the selected curve and its position and u-v parameters.

.NET Syntax

Visual Basic (Declaration)

```
Function IFindMinimumRadius( _  
    ByRef Bound As System.Double, _  
    ByRef NumOfRadius As System.Integer, _  
    ByRef Radius As System.Object, _  
    ByRef Location As System.Object, _  
    ByRef Parameter As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ICurve  
Dim Bound As System.Double  
Dim NumOfRadius As System.Integer  
Dim Radius As System.Object  
Dim Location As System.Object
```



```
Dim Parameter As System.Object
Dim value As System.Boolean

value = instance.IFindMinimumRadius(Bound, NumOfRadius, Radius, Location, Parameter)
```

C#

```
System.bool IFindMinimumRadius(
    ref System.double Bound,
    out System.int NumOfRadius,
    out System.object Radius,
    out System.object Location,
    out System.object Parameter
)
```

C++/CLI

```
System.bool IFindMinimumRadius(
    & System.double% Bound,
    & [Out] System.int NumOfRadius,
    & [Out] System.Object^ Radius,
    & [Out] System.Object^ Location,
    & [Out] System.Object^ Parameter
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bound
Array containing the start and end boundary parameters (see **Remarks**)

NumOfRadius
Number of radius returned; can be 0 or 1

Radius
Minimum radius of curvature (see **Remarks**)

Location
[Position](#) where minimum radius curvature occurred (see **Remarks**)

Parameter
Curve parameter (see **Remarks**)

Return Value

True if operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [Curve::IFindMinimumRadius](#).

Example

Unmanaged C++ COM:

```
////////////////////////////////////
// IMinRadius implementation
STDMETHODIMP CMinRadius::StartNotepad()
{
    // TODO: Add your implementation code here
    CComPtr<IModelDoc> iModelDoc2;
    m_islWorks->get_IActiveDoc(&iModelDoc2);

    CComPtr<ISelectionMgr> swSelMgr;
    iModelDoc2->get_ISelectionManager(&swSelMgr);
    struct IDispatch* pDispSelection;
    CComPtr<IEdge> swEdge;
    swSelMgr->GetSelectedObject6(1, -1, &pDispSelection);
    pDispSelection->QueryInterface(
```

```

__uuidof(IEdge), reinterpret_cast<void**>(&swEdge));
pDispSelection->Release();

double boundArr[2];
CComPtr<ICurve> swCurve;
swEdge->IGetCurve(&swCurve);

VARIANT_BOOL isClose, isPeriodic, testRetVal;
swCurve->GetEndParams(&boundArr[0], &boundArr[1], &isClose, &isPeriodic, &testRetVal);

long numOfRadius(0);
VARIANT radius, Location, UVParameter;
VARIANT_BOOL status(FALSE);
swCurve->IFindMinimumRadius(boundArr, &numOfRadius, &radius, &Location, &UVParameter, &status);

SafeDoubleArray radiusSA(radius);
double test1 = radiusSA[0];

SafeDISPATCHArray locationSA(Location);

CComPtr<IMathPoint> swMathPoint1;
locationSA[0]->QueryInterface(
__uuidof(IMathPoint), reinterpret_cast<void**>(&swMathPoint1));

double locationArra1[3];

swMathPoint1->get_IArrayData(locationArra1);

SafeDoubleArray UVParameterSA(UVParameter);

double uvpara1 = UVParameterSA[0];
return S_OK;
}

```

Remarks

The search is confined to the portion of the selected curve lying inside of Bound.

COM returns these data types for these parameters:

- Radius: VARIANT of SafeDoubleArray
- Location: VARIANT of SafeDispatchArray of [IMathpoint](#)
- Parameter: VARIANT of SafeDoubleArray

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::FindMinimumRadius Method](#)
[ISurface::FindMinimumRadius Method](#)
[ISurface::IFindMinimumRadius Method](#)
[ICurve::GetEndParams Method](#)

Availability

IGetBCurveParams Method (ICurve)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IGetBCurveParams Method (ICurve)

Obsolete. Superseded by [ICurve::IGetBCurveParams3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBCurveParams() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurve
Dim value As System.Double

value = instance.IGetBCurveParams()
```

C#

```
System.double IGetBCurveParams()
```

C++/CLI

```
System.double IGetBCurveParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to an array of doubles describing the parameters of the curve (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Curve::IGetBCurveParams](#).

Example

[Get Curve Spline Points \(C++\)](#)

Remarks

Call [ICurve::IGetBCurveParamsSize2](#) to determine the size of the array returned and whether the curve data returned by [ICurve::IGetBCurveParams](#) is cubic rational or not.

To control the accuracy of the curve data, see [IModeler::SetToleranceValue](#).

The return value is an array of doubles. The array size is (2 + numKnots + numControlPointDoubles) where numKnots is (numControlPoints + order). The array is as follows:

```
[ packedDouble1, packedDouble2, knot1, knot2,..., ControlPoint1[Dimension], ControlPoint2[Dimension],...
]
```

where:

- *packedDouble1* is an integer pair containing the *Dimension* and *Order*
- *packedDouble2* is an integer pair containing the *NumControlPoints* and *Periodicity*
- *knot1*
- *knot2*

...

- *ControlPoint1[dimension]*
- *ControlPoint2[dimension]*

...

The size of the control point array is based on the curve dimension:

- If *Dimension* = 3, then *ControlPoint* is an array of 3 doubles (x, y, z)
- If *Dimension* = 4, then *ControlPoint* is an array of 4 doubles (x, y, z, w) where w = weight

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::GetBCurveParams Method](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::SimplifyBcurve Method](#)
[ICurve::IConvertPcurveToBcurveSize Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::IConvertArcToBcurveSize Method](#)
[ICurve::IConvertLineToBcurveSize Method](#)
[ICurve::MakeBsplineCurve Method](#)

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IGetBCurveParams3 Method (ICurve)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [ICurve Interface](#) : IGetBCurveParams3 Method (ICurve)

Obsolete. Superseded by [ICurve::GetBCurveParams4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBCurveParams3( _
    ByVal ArraySize As System.Integer _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurve
Dim ArraySize As System.Integer
Dim value As System.Double

value = instance.IGetBCurveParams3(ArraySize)
```

C#

```
System.double IGetBCurveParams3(
    System.int ArraySize
)
```

C++/CLI

```
System.double IGetBCurveParams3(
    & System.int ArraySize
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArraySize
Size of the return array

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles describing the parameters of the curve
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Use [ICurve::IGetBCurveParamsSize3](#) to determine the size of the array returned and whether the curve data returned by Curve::IGetBCurveParams3 is cubic rational or not.

To control the accuracy of the curve data, see [IModeler::SetToleranceValue](#).

The return value retval is an array of doubles. The array size is (2 + numKnots + numControlPointDoubles) where numKnots is (numControlPoints + order). The array is as follows:

[*packedDouble1*, *packedDouble2*, *knot1*, *knot2*,..., *ControlPoint1[Dimension]*, *ControlPoint2[Dimension]*,...]]

where:

- *packedDouble1* is an integer pair containing the *Dimension* and *Order*
- *packedDouble2* is an integer pair containing the *NumControlPoints* and *Periodicity*

- *knot1*
- *knot2*
- ...
- *ControlPoint1[dimension]*
- *ControlPoint2[dimension]*
- ...

The size of the control point array is based on the curve dimension:

- If *Dimension* = 3, then *ControlPoint* is an array of 3 doubles (x, y, z)
- If *Dimension* = 4, then *ControlPoint* is an array of 4 doubles (x, y, z, w) where w = weight

See Also

[ICurve Interface](#)
[ICurve Members](#)
[ICurve::GetBCurveParams3 Method](#)
[ICoEdge::GetCurveParams Method](#)
[ICoEdge::IGetCurveParams Method](#)
[ICurve::CircleParams Property](#)
[ICurve::ICircleParams Property](#)
[ICurve::ILineParams Property](#)
[ICurve::LineParams Property](#)
[ICurve::ConvertArcToBcurve Method](#)
[ICurve::ConvertLineToBcurve Method](#)
[ICurve::GetEllipseParams Method](#)
[ICurve::GetPCurveParams Method](#)
[ICurve::IGetPCurveParams Method](#)
[ICurve::IGetEllipseParams Method](#)
[ICurve::Identity Method](#)
[ICurve::IGetSplinePts Method](#)
[ICurve::IsBcurve Method](#)
[ICurve::IsCircle Method](#)
[ICurve::IsEllipse Method](#)
[ICurve::IsLine Method](#)
[ICurve::MakeBsplineCurve2 Method](#)
[IEdge::GetCurveParams2 Method](#)
[IEdge::IGetCurveParams2 Method](#)
[IModeler::CreatePCurve Method](#)
[IModeler::ICreatePCurve Method](#)

Availability

SOLIDWORKS 2007 SP5, Revision Number 15.5

SOLIDWORKS API Help

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IGetBCurveParamsSize Method (ICurve)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurve Interface](#) : IGetBCurveParamsSize Method (ICurve)

Obsolete. Superseded by [ICurve::IGetBCurveParamsSize2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBCurveParamsSize( _
    ByVal WantCubicIn As System.Boolean _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurve
Dim WantCubicIn As System.Boolean
Dim value As System.Integer
```

```
value = instance.IGetBCurveParamsSize(WantCubicIn)
```

C#

```
System.int IGetBCurveParamsSize(
    System.bool WantCubicIn
)
```

C++/CLI

```
System.int IGetBCurveParamsSize(  
& System.bool WantCubicIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubicIn

Visual Basic for Applications (VBA) Syntax

See [Curve::IGetBCurveParamsSize](#).

See Also

[ICurve Interface](#)
[ICurve Members](#)